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A BASIS RESULT IN COMBINATORY LOGIC

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The aim of this article is to show that a basis for combinatory logic [2] must contain at least one combinator with rank strictly greater than two. We use notation of [1].

DEFINITION 1. Let Q be a primitive combinator given by its reduction rule $Qx_1 \cdots x_n \rightarrow C$, where C is a pure combination of the variables $x_1, \dots, x_n$. n is called the *rank* of the combinator.

DEFINITION 2. A set $\{Q_1, \dots, Q_n\}$ of combinators is a *basis for combinatory logic* if for every finite set $\{x_1, \dots, x_m\}$ of variables and every pure combination C of these variables, there exists a pure combinator Q of $Q_1, \dots, Q_n$ such that $Qx_1 \cdots x_m \rightarrow C$.

Property. The *Church-Rosser theorem* and the *(quasi-)normalization theorem* are valid for the combinatory reduction system under consideration.

Proof. See [4], [5], and [6].

THEOREM. Any basis must contain at least one combinator with rank strictly greater than two.

Let us assume a basis $\mathbf{B}$ with combinators of rank strictly less than 3; then there exists a pure combination X of the combinators in $\mathbf{B}$ such that: $XABC \rightarrow CAB$.

(*) First of all, we remark that if $XABC \rightarrow M \rightarrow CAB$, M must contain at least one occurrence of each of the variables A , B and C .

Notation. We denote by $E[X_1, \dots, X_n]$ expressions that contain at least one occurrence of every term $X_1, \dots, X_n$.

LEMMA 1. Head reduction of XAB leads to a term $T(E_1[A, B])$, where T is a combinator. Therefore, $XABC$ reduces to a term $E_2[E_1[A, B], C] \equiv T(E_1[A, B])C$.

PROOF. If the head reduction of XAB is infinite, then also the head reduction of $XABC$ is infinite. Hence, by the normalization theorem, $XABC$ cannot have a normal form. Contradiction.

Head reduction of XAB will stop in three possible cases:

- 1) $XAB \rightarrow T(E_1[A, B])$ and T is a combinator with rank 2; or
- 2) $XAB \rightarrow U$ and U is a combinator (this case is impossible because of (*)); or
- 3) $XAB \rightarrow V_1 \cdots V_n$ with V_1 a variable (i.e. A or B). But V_1 will stay in head position during all further contractions of $XABC$ until the normal form CAB is obtained. This is impossible because V_1 is different from C .

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Therefore, we obtain $XABC \rightarrow T(E_1[A, B])C \equiv E_2[E_1[A, B], C]$.

DEFINITION. By a π -expression we mean any expression M where all occurrences of variables A and B are in occurrences of a subterm $E_1[A, B]$ containing no occurrence of C and such that M contains at least one occurrence of C . Such expressions can be denoted by $E_2[E_1[A, B], C]$.

LEMMA 2. Let M be a π -expression such that M reduces to CAB . Then there exists a π -expression N which reduces to CAB such that $M \rightarrow N$ with at least one leftmost-contraction.

PROOF. When we perform the leftmost contraction of a π -expression $E_2[E_1[A, B], C]$, three cases are possible:

1) The head combinator of the redex is not in $E_1[A, B]$. Thus, leftmost contraction conserves the status of π -expression because $E_1[A, B]$ can only be an argument of the head combinator. Some occurrences may disappear, but at least one of them must exist since A and B occur in the normal form.

2) The leftmost redex is in $E_1[A, B]$. Then $E_1[A, B]$ will reduce to another expression $E_3[A, B]$, and $E_2[E_1[A, B], C] \rightarrow E_2[E_3[A, B], C]$ is obtained by the contraction of all occurrences of $E_1[A, B]$. It is still a π -expression.

3) The head combinator of the redex is in $E_1[A, B]$, but the redex is not. Let H be this combinator; H must have rank 2 and $E_1[A, B] \equiv H(E_3[A, B])$. Therefore, the original term $E_2[E_1[A, B], C]$ must be rewritten $E_2[H(E_3[A, B]), C]$ and then $E_4[E_3[A, B], C]$ so that the head combinator is no longer in $E_3[A, B]$ and case 1) applies.

The term N reduces to CAB : this is a consequence of the Church-Rosser theorem.

PROOF OF THE THEOREM. We define a reduction sequence of $XABC$ as follows:

$$XABC \rightarrow E_2[E_1[A, B], C] \quad \text{by Lemma 1.}$$

Then apply the preceding Lemma 2 so that

$$XABC \rightarrow E_2[E_1[A, B], C] \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow \dots$$

and each term M_{i+1} is a π -expression obtained from the π -expression M_i by the reduction of Lemma 2. We must remark that the strategy is quasi-leftmost since each step of reduction is concerned with at least one leftmost contraction. Because of the (quasi)-normalization theorem [3], this strategy must lead to the normal form. The contradiction arises because (CAB) is not a π -expression.

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REFERENCES

- [1] H. P. BARENDREGT, *The lambda-calculus: its syntax and semantics*, North-Holland, Amsterdam, 1981.
- [2] P. BELLOT, *A new proof for Craig's theorem*, this JOURNAL, vol. 50 (1985), pp. 395–396.
- [3] J. R. HINDLEY and J. P. SELDIN, *Introduction to combinators and lambda-calculus*, Cambridge University Press, Cambridge, 1986.

- [4] G. HUET, *Confluent reductions: abstract properties and applications to term rewriting systems*, *Journal of the Association for Computing Machinery*, vol. 27 (1980), pp. 797–821.
- [5] J. W. KLOP, *Combinatory reduction systems*, Mathematical Centre Tracts, vol. 127, Mathematisch Centrum, Amsterdam, 1980.
- [6] M. J. O'DONNELL, *Computing in systems described by equations*, Lecture Notes in Computer Science, vol. 58, Springer-Verlag, Berlin, 1977.

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